

A Hybrid Machine Learning Approach to Identifying Genetic Signatures of Pediatric Crohn's Disease from RNA-Sequencing Data

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Abstract

This project used a machine-learning approach to analyze RNA sequencing data obtained from pediatric individuals diagnosed with Crohn's disease (CD), alongside a control group without an inflammatory bowel disease (IBD). The goal was to identify the most influential genes in classifying between these groups, and to offer insights into the genetic landscape of pediatric CD. A hybrid machine learning method was implemented, where a random forest (RF) classifier was used as a feature selection step to find the set of genes that yielded the highest accuracy in neural networking (NN) training. This optimal set of features consisted of 275 genes, which after conducting a pathway analysis, were found to be highly involved in the immune system, particularly the functioning of granulocytes. The top upstream regulators of the group of genes were also all molecules highly linked to cytokine signaling and immune response regulation. These findings illustrate that the immunological landscape, particularly involving granulocyte activity and cytokines signaling, are important signatures of pediatric CD. Training the neural network with the cohort of 275 selected genes was highly successful, with an accuracy of 0.97 in classifying between the healthy and disease states. However, it is noteworthy that when only the top one gene was selected from the RF classifier to be used in NN training, the accuracy had a relatively high score of 0.88. This gene was PELATON (LINC01272), which is a long non-coding RNA (lncRNA) that has previously been found to be upregulated in pediatric CD (Haberman et al., 2018). This finding points to the need for long non-coding RNAs to be further studied in relation to CD as they hold significant potential as therapeutic targets with less off-target toxic effects than protein-coding RNAs.

Introduction

Crohn's disease is an inflammatory bowel disease, characterized by chronic inflammation and dysregulation of the gastrointestinal (GI) tract. While pediatric cases account for only 10-25% of Crohn's instances, the disease's severity and prognosis are worse in children than in adults. Pediatric CD also has a stronger genetic underpinning, with less influence from environmental factors in its pathogenesis (Imielinski et al., 2009). GWAS studies have revealed many genetic loci that are correlated with pediatric CD, many of which are linked to immune pathways and cytokine signaling, DNA repair, oxidative phosphorylation, and extracellular matrix (ECM) pathways (Chen et al., 2024). Long non-coding RNAs (lncRNAs), which are important for regulating transcription, have also been linked to pediatric CD. Haberman et. al. identified HNF4A-AS1 and LINC01272 as two lncRNAs that are particularly significant, and both are part of a large network of co-expressed genes (Haberman et al., 2018). Machine learning (ML) has been used to help identify genes and pathways of interest in Crohn's disease, but typically logistic regression or other traditional ML methods are used. This study instead uses a combined random forest (RF) classifier and neural network (NN) approach to identifying genes and pathways of interest from RNA-sequencing data. The RF classifier is used to reduce the dimensionality of complex, noisy RNA-seq data and find the optimal set of genes for NN training, and the NN is used because of its demonstrated ability to capture non-linear relationships from genetic data. Based on previous research, it is expected that genes highly linked to immune functioning or lncRNAs will strongly influence prediction.

Methods

The RNA-seq dataset was obtained from the RISK study, which collected data from ileal biopsies from pediatric Crohn's patients across North America who were all treatment-naïve. The RNA sequencing dataset consisted of 245 samples, 210 of which were from pediatric Crohn's treatment-naïve patients, and a group of 35 controls who did not have an inflammatory bowel disease. The dataset consisted of raw counts from 65,217 genes. This data was first normalized using a counts per million (CPM) approach and then scaled down to a range of 0-1 using MinMaxScaler().

To reduce the number of features that the NN would be trained on, a feature reduction approach using a RF classifier was implemented. The RF classifier uses an ensemble of decision trees to perform classification. Once trained, the most important genes in classification can be directly extracted using feature importance. Cross-validation was used to select the optimal number of genes from the RF classifier to use in training the neural network.

The NN was then trained with this optimal group of genes in the feature set. The labels were one hot encoded into two classes, the control and Crohn's groups. The data was then divided into training and testing sets, with sizes 0.8 and 0.2, for training and evaluation purposes. The NN consisted of one dense layer with 64 neurons. Only one layer was used since adding

additional layers to the model had no significant impact on accuracy. The model used Adam optimizer and binary cross-entropy loss, and was trained for 50 epochs. Further pathway analysis was conducted with Ingenuity Pathway Analysis (IPA) to determine the top canonical pathways and upstream regulators implicated by the genes that the neural network was trained on.

Results and Discussion

Figure 1 shows that the highest accuracy, which was 0.97, was achieved when the top 275 genes that were most influential in the RF Classifier were extracted for the model to be trained on. However, it is noteworthy that when only the one most important gene from the RF classifier was used, the neural network achieved an accuracy of 0.88. This gene was PELATON, which is a lncRNA also known as LINC01272. This gene was also identified by Haberman et. al. as being a highly upregulated gene in pediatric CD. LINC01272 is highly co-expressed with other genes, which might explain why it was sufficient for the model to make predictions with relatively high accuracy. Non-coding RNAs are not as well studied as protein-coding RNAs in relation to CD, but they show more promise as therapeutic targets because they often have less off-target toxic effects. Since LINC01272 was highly predictive of pediatric CD in this study, it should be further investigated as a potential therapeutic target.

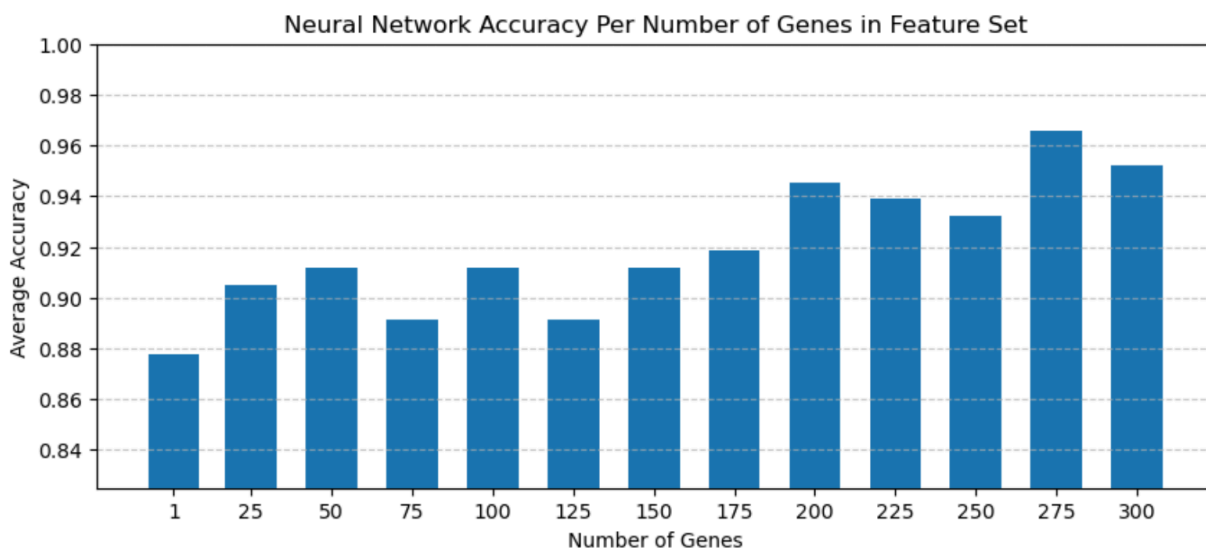


Figure 1 Number of Genes vs. NN Accuracy.

After training the model on the selected 275 genes and evaluating it, the accuracy was 0.97 and the loss was 0.10. The model's accuracy and loss over time are shown in Figures 2a and 2b, respectively. Additionally, the F1 score, a metric that takes both precision and recall into account, was 0.99 for the Crohn's group and 0.92 for the controls. This means that the model was slightly better at correctly predicting the disease group, which is likely because it had a larger

sample size than the control. Overall, the model was proficient with a high accuracy and predictive ability, showing that the 275 genes selected from the random forest method were successful in capturing the differing genetic profiles between the groups.

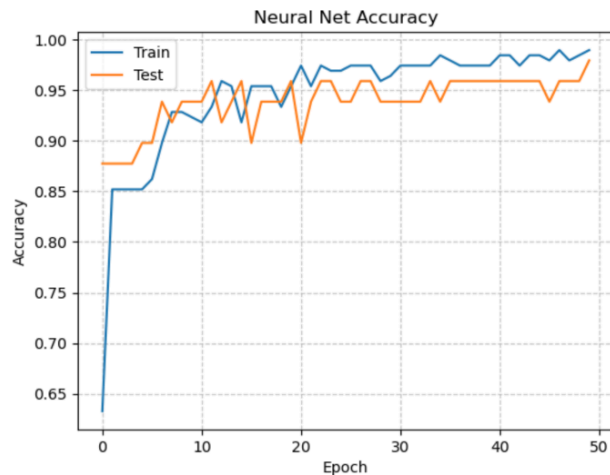


Figure 2a Neural Net Accuracy

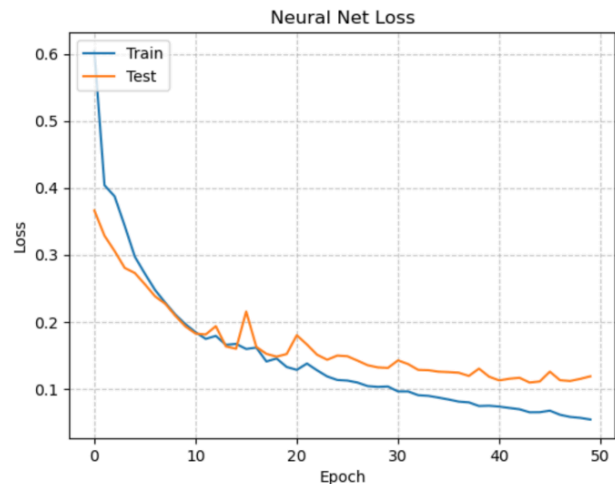


Figure 2b Neural Net Loss

An analysis was run using Ingenuity Pathway Analysis (IPA) to find the top pathways and upstream regulators of these genes. As shown in Table 1, the top pathways affected are neutrophil degranulation, granulocyte adhesion and diapedesis, the complement system, the osteoarthritis pathway, and G alpha (i) signaling events. The first three are closely related and involve the immune system response, especially the function of granulocytes and neutrophils, which are known to be disrupted in pediatric CD. Previous research has demonstrated that impaired neutrophil function increases the risk of developing inflammatory bowel disease and that in CD there is a lack of neutrophil recruitment in the immune response (Levine et al., 2013). The G alpha (i) signaling pathway was also found by IPA to be significant, which might be due to its role in regulating immune responses. The osteoarthritis pathway was also highlighted as a top pathway, which was unexpected but can indicate potential shared genetic loci between early-onset Crohn's and osteoarthritis.

Table 1 Top Pathways from IPA Analysis

Name	P-Value
Neutrophil degranulation	3.24E-08
Granulocyte Adhesion and Diapedesis	1.38E-05
Complement System	4.06E-04
Osteoarthritis Pathway	4.41E-04
G alpha (i) signalling events	8.34E-04

Table 2 Top Upstream Regulators from IPA Analysis

Name	P-Value
Lipopolysaccharide	3.43E-12
TNF	1.72E-10
NFkB (complex)	1.56E-08
IL17RA	3.76E-08
Peptidoglycan	4.22E-08

Table 2 shows the top upstream regulators identified by the IPA analysis of the 275 genes used in NN training. The molecules found play key roles in the immune response and are involved in cytokine signaling. Since immune regulation and cytokine signaling disruption are hallmarks of CD, this finding is expected. In particular, Imielinski et al previously found that the T-helper 17 cell pathway was significant in the pathogenesis of early-onset Crohn's (2009), which aligns with IL17RA (Interleukin 17 Receptor A) being a top upstream regulator since it binds to IL-17, a cytokine secreted by T-helper 17 cells. More research is needed to determine more specifically how these molecules contribute to the pathogenesis of pediatric CD, which could lead to earlier interventions and improved treatment strategies for the disease. However, a limitation posed by this study is that the sample size was small, and the control group was also significantly smaller than the CD group. Furthermore, only one neural network model was explored, whereas other types such as an LSTM or CNN might yield different results. Further studies would benefit from a larger sample size and the exploration of differing deep-learning architectures in the pursuit of genetic signatures of pediatric CD.

Conclusion

In summary, the study successfully identifies genetic signatures of pediatric Crohn's disease (CD) through a hybrid machine learning approach that can classify between a healthy and disease state with high accuracy. Using a combination of RF feature selection and NN classification, 275 genes were found that are important for distinguishing CD patients from controls, achieving an accuracy of 0.97. Surprisingly, even with just one gene, PELATON (LINC01272), the model demonstrates significant predictive power, illustrating its potential as a therapeutic target. Pathway analysis also revealed insights into the underlying mechanisms of Crohn's disease (CD) in pediatric cases. These findings highlight the significance of investigating both non-coding RNAs and immune pathways, particularly those involving granulocytes and cytokine signaling, in the pathology of the disease. By identifying key immune-related pathways and upstream regulators, researchers can better understand the disease's pathogenesis and potentially identify targets for therapeutic intervention. Moreover, the study demonstrates the value of integrating neural networks in analyzing genomic data. Moving forward, larger datasets and exploration of alternative deep learning architectures could provide new insights into the genetic landscape of pediatric CD and inform the development of more effective and personalized treatment strategies.

References

- Chen, K.A., Nishiyama, N.C., Kennedy Ng, M.M. et al. (2024). Linking gene expression to clinical outcomes in pediatric Crohn's disease using machine learning. *Sci Rep* 14, 2667. <https://doi.org/10.1038/s41598-024-52678-0>
- Haberman, Y., BenShoshan, M., Di Segni, A., Dexheimer, P. J., Braun, T., Weiss, B., Walters, T. D., Baldassano, R. N., Noe, J. D., Markowitz, J., Rosh, J., Heyman, M. B., Griffiths, A. M., Crandall, W. V., Mack, D. R., Baker, S. S., Kellermayer, R., Patel, A., Otley, A., Steiner, S. J., Denson, L. A. (2018). Long ncRNA Landscape in the Ileum of Treatment-Naive Early-Onset Crohn Disease. *Inflammatory bowel diseases*, 24(2), 346–360. <https://doi.org/10.1093/ibd/izx013>
- Imielinski, M., Baldassano, R. N., Griffiths, A., Russell, R. K., Annese, V., Dubinsky, M., Kugathasan, S., Bradfield, J. P., Walters, T. D., Sleiman, P., Kim, C. E., Muise, A., Wang, K., Glessner, J. T., Saeed, S., Zhang, H., Frackelton, E. C., Hou, C., Flory, J. H., Otieno, G., Hakonarson, H. (2009). Common variants at five new loci associated with early-onset inflammatory bowel disease. *Nature genetics*, 41(12), 1335–1340. <https://doi.org/10.1038/ng.489>
- Kong, Y., Yu, T. (2018). A Deep Neural Network Model using Random Forest to Extract Feature Representation for Gene Expression Data Classification. *Sci Rep* 8, 16477. <https://doi.org/10.1038/s41598-018-34833-6>
- Levine, A. P., & Segal, A. W. (2013). What is wrong with granulocytes in inflammatory bowel diseases? *Digestive diseases* (Basel, Switzerland), 31(3-4), 321–327. <https://doi.org/10.1159/000354686>
- Pigneur B, Seksik P, Viola S, Viala J, Beaugerie L, Girardet JP, Ruemmele FM, Cosnes J. (2010). Natural history of Crohn's disease: comparison between childhood- and adult-onset disease. *Inflamm Bowel Dis*. (6):953-61. doi: 10.1002/ibd.21152. PMID: 19834970.